

4. (a) Theories about the Universe have changed over time.

Tick (✓) the boxes next to the names of **two** of these theories.

[2]

Liquid State

☐

Steady State

☐

Loud Bang

☐

Big Bang

☐

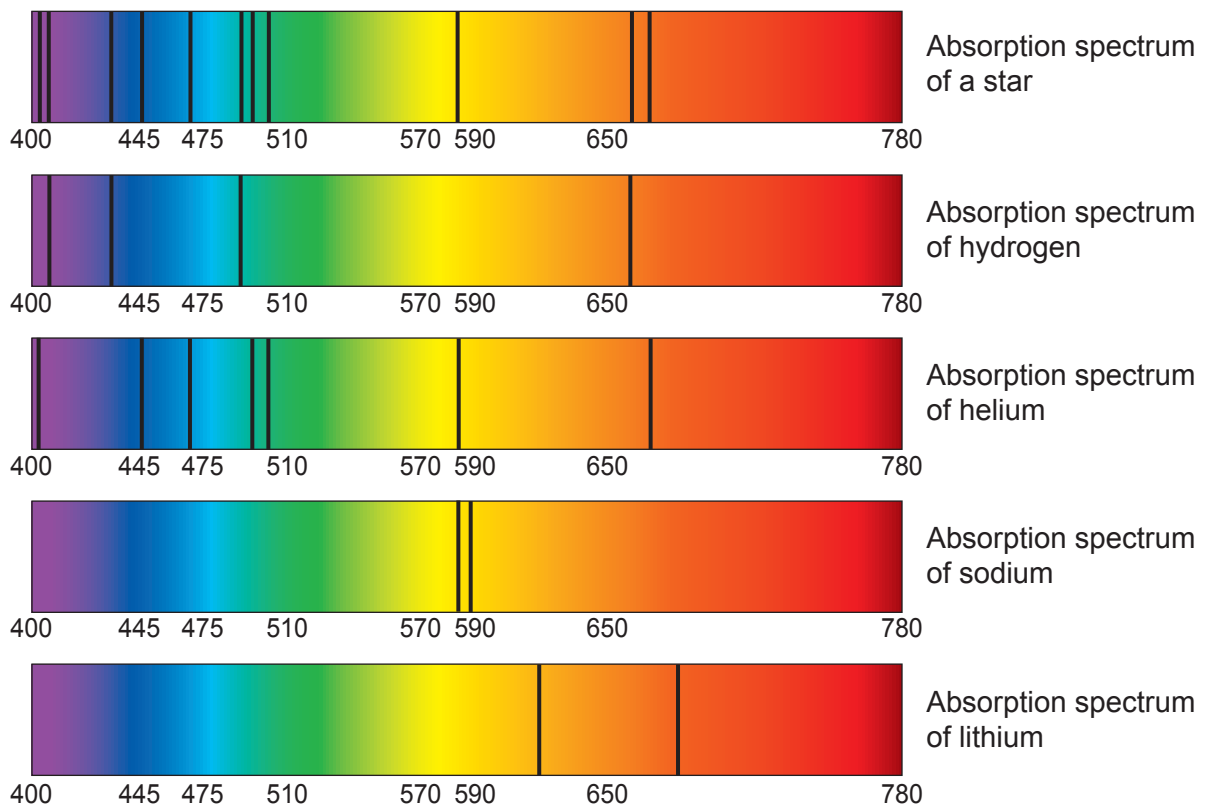
- (b) Images of the Universe are taken using electromagnetic waves.

Name **two** types of electromagnetic wave.

[2]

..... and .....

- (c) The diagram shows a spectrum from a star and four elements.



Examiner  
only

Complete the table by placing a tick (✓) if the element is present in the star. Hydrogen has been done for you.

[1]

| Element  | Element is present (✓) |
|----------|------------------------|
| hydrogen | ✓                      |
| helium   |                        |
| sodium   |                        |
| lithium  |                        |

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(d) The table gives information about the planets in our solar system.

|                                         | Mercury | Venus | Earth | Mars | Jupiter | Saturn | Uranus | Neptune |
|-----------------------------------------|---------|-------|-------|------|---------|--------|--------|---------|
| density ( $\text{kg/m}^3$ )             | 5427    | 5243  | 5514  | 3933 | 1326    | 687    | 1271   | 1638    |
| mean distance from the Sun (million km) | 58      | 108   | 150   | 228  | 779     | 1434   | 2873   | 4495    |
| mean speed of orbit (km/s)              | 47.4    | 35.0  | 29.8  | 24.1 | 13.1    | 9.7    | 6.8    | 5.4     |
| mean temperature ( $^{\circ}\text{C}$ ) | 167     | 464   | 15    | -65  | -110    | -140   | -195   | -200    |
| number of moons                         | 0       | 0     | 1     | 2    | 79      | 82     | 27     | 14      |

Use the information in the table to answer the questions below.

(i) Name the coldest **rocky** planet. .... [1]

(ii) State how many **rocky** planets have moons. .... [1]

(iii) Name the gas giant that orbits the Sun with the greatest **speed**. .... [1]

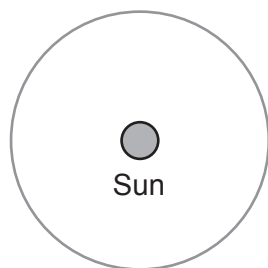
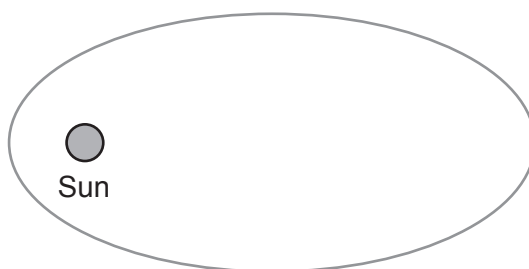
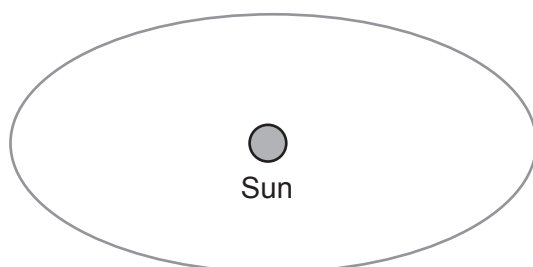
(iv) Eris is a dwarf planet. Its orbit is 10 180 million km away from the Sun.  
Shaun says it is found in the asteroid belt between the orbits of Mars and Jupiter.  
Explain whether you agree with Shaun. [1]

.....  
.....  
.....



(e) Tick (✓) the box next to the shape of the orbit of a comet around the Sun.

[1]

☐☐☐

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